Capstone Project

Instructions

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Objective

The objective of this module is to have you **create a complete AI project** so that you can put into practice the knowledge acquired throughout your university curriculum. This is not a simple tutorial/practical work: you will be **almost entirely autonomous**, and your project will have to **solve a complex problem by implementing a complete AI life cycle**.

The pedagogical objectives are as follows:

- Apply **Machine Learning/Deep Learning techniques**.
- Manage a **data pipeline**.
- **Work as a team**.
- **Communicate** technical results.

Subject Search

The first step is to define a project. We expect it to **be feasible**—that is, not too ambitious—and **not too simple**. *For example, applying Random Forest to the Titanic dataset is much too simple and will be refused as a project.*

Note that the complexity of the machine learning techniques used is not everything. Indeed, a project incorporating advanced deep learning techniques but used online as part of tutorials is not acceptable. *For example, a RAG Chatbot project for legal code or technical documents, although complex, is too simple.*

Generally, we will evaluate the quality of your project based on the **problem to be solved**, the **necessary data**, and the **machine learning techniques used**.

Example Subject

Autonomous AI Temporary Employment Agency

*The goal is not to have a conversation with the AI, but to give it a global objective—**such as organizing interviews with the 3 best candidates for a position**—and to let it act autonomously.*

*As input, the AI system could receive files describing the position as well as all submitted applications (CVs, cover letters, etc.). The output would be scheduling appointments with the 3 best candidates, accompanied by a summary explaining why they were chosen and technical questions to ask them during the interview, based on their profiles. **All of this without any human intervention**.*

*To achieve this, **at least 4 agents are used**: an orchestrator agent (which manages the task, delegates to other agents, and verifies the result), an extractor agent (which reads the CVs, letters, etc.), an evaluator agent (which checks the candidates' compatibility with the job posting), and a communication agent (which manages emails via Gmail, appointments in Google Calendar, etc.).*

Data

Today, downloading an excellent model only requires three lines of code. The **competitive advantage therefore mainly lies in the data,** especially since most of an AI engineer's time is dedicated to searching, cleaning, and preparing data.

It is strictly forbidden to use "toy" datasets (Titanic, Iris, MNIST, Housing, etc.) **or datasets that are already perfect** (without missing values, without noise, etc.).

You **must justify the source, quality, and legality of your data**. A dataset that you have collected and cleaned yourself will be valued over a pre-existing dataset.

Data

Data can be found everywhere: on specialized sites, with public APIs, by scraping websites, by generating it with AI, or by collecting it yourself... Don't hesitate to combine sources.

kaggle

datagouv

Datasets

Google Dataset Search

aws open data

Data

To check if your data is valid, ask yourself the following questions:

Do I have enough examples?

Do I have the labels?
If I have to label them manually, how long will it take?

Is my data (labels) balanced?

Are there a lot of missing values or errors?

In any case, remember to present your data to us at the same time as your project so that we can validate it.

Methodology

From a purely technical standpoint, we expect you to use Python as the programming language. You are free to choose the libraries you wish to use. Note that the code part of your project is important, but it does not represent the entirety of it. Indeed, a simplified MLOps pipeline can be represented by the following sequence:

Problem Definition

→ Data Collection

→ Cleaning

→ Modeling

→ Evaluation

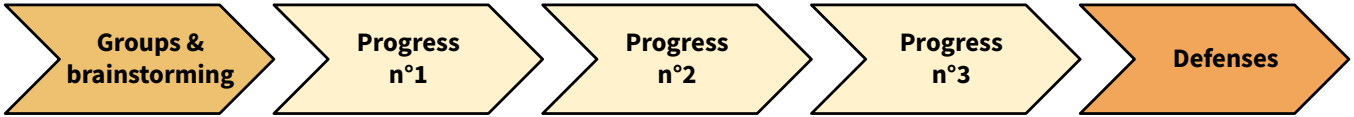
→ Deployment

All of these phases must appear, be documented, analyzed, and justified in your project.

Group and Milestones

You will complete your project in **groups of 3 with**, at minimum, 1 person more "business"-oriented and 1 person more "technical"-oriented.

This first session is dedicated to forming groups and brainstorming your project. You will then have **three checkpoints** to monitor the progress of your project. These checkpoints will not only help you with your project but will also be used for your evaluation. In fact, **each of these controls is part of the final grade**. We do not expect you to have finished the project from the very first session, only that you have advanced enough to not be late and avoid doing everything at the last minute. A **progress report** is expected at each checkpoint. The last session is dedicated to defenses/presentations.



Expected Deliverables

The expected deliverables are as follows:

Source code (clean, commented, on GitHub).

Technical report or Blog article.

Oral presentation with Demo.

All deliverables are expected on the day of the defense/presentation at midnight at the latest.

Oral Presentation

The oral presentation must **last between 10 and 15 minutes**. The students in a group must present the entire project equitably. It is **forbidden to present only the introduction or the conclusion**, especially if these are not very informative.

Grades will be awarded to both the group and individually. We reserve the right to reduce the grade of a group member if they do not master their project.

A **5 to 10-minute question-and-answer session** may be added at the end of the presentation.

Evaluation Rubric

Oral Defense & Demo – ***40% of the final grade***

- Problem Resolution
- Storytelling & Explaining
  - *Ability to engage the audience + the problem is explained simply*
- The Demo
  - *The demo works live (No pre-recorded video, except for a catastrophic plan B).*
  - *Clear demo scenario.*
- Technical Mastery
  - *Precise answers to the jury's questions.*
  - *Ability to justify technical choices.*
- Visual Aids and Time Management
  - *No copy-pasting of illegible code on the slides.*
  - *Clean slides, little text, many visuals.*

Evaluation Rubric

Technical Report – ***40% of the final grade***

- Problem Framing & Business Value
  - *The problem is clearly defined.*
  - *Definition of success metrics*
- The Data
  - *Explanation of collection and cleaning.*
  - *Relevant graphs showing distributions, correlations, and potential biases.*
  - *Justification of preprocessing choices*
- Modeling & Experimentation
  - *Comparison with a Baseline*
  - *Justification of the choice of algorithms.*
  - *Error analysis: showing where the model makes mistakes and why.*
- Interpretability & Ethics
  - *Attempt to explain predictions*
  - *Discussion on the model's limitations, ethical biases, and carbon impact.*

Evaluation Rubric

Source Code & Engineering – ***20% of the final grade***

- Reproducibility & Installation
  - *The README.md is impeccable (title, description, how to install, how to run).*
  - *Presence of a functional requirements.txt or environment.yml.*
  - *The project launches without error on a third-party machine (ideally Dockerized).*
- Architecture & Structure
  - *The training and inference code is moved out of notebooks and organized into modular Python scripts (.py in folders like src/, data/, models/).*
  - *Clear separation between pre-processing, training, and the API.*
- Code Quality
  - *The code is intelligently commented.*
  - *Error management (try/except) for the APIs + adherence to conventions.*